

 CDM Project Activity Registration and Validation Report Form (By submitting this form, designated operational entity confirms that the proposed CDM project activity meets all validation and registration requirements and thereby requests its registration)	
Section 1: Request for registration	
Name of the designated operational entity (DOE) submitting this form	SGS United Kingdom Ltd.
Title of the proposed CDM project activity (Section A.2 of the attached CDM-PDD) submitted for registration	Project for GHG emission reduction by thermal oxidation of HFC 23 in Gujarat, India.
Project participants (Name(s))	▪ Cooperatieve Centrale Raiffeisen Boerenleenbank B.A. (Rabobank), Netherlands ▪ Gujarat Fluorochemicals Limited, Gujarat, India ▪ Ineos Fluor Limited, United Kingdom ▪ Sumitomo Corporation, Japan
Sector in which project activity falls	Fugitive emissions from production and consumption of halocarbons and sulphur hexafluoride
Is the proposed project activity a small-scale activity?	<u>No</u> (underline as applicable)
Section 2: Validation report	
List of documents to be attached to this validation report (please check mark):	
☒ The CDM-PDD of the project activity ☒ An explanation by the submitting designated operational entity of how it has taken due account of comments on validation requirements received, in accordance with the CDM modalities and procedures, from Parties, stakeholders and UNFCCC accredited non-governmental organizations; ☒ The written approval of voluntary participation from the designated national authority of each Party involved, including confirmation by the host Party that the project activity assists it in achieving sustainable development: ☒ (Attach a list of all Parties involved and attach the approval (in alphabetical order)) Host Party: ☐ India Other Parties involved: ☐ Japan ☐ Netherlands ☐ United Kingdom of Great Britain and Northern Ireland ☒ Other documents, including any validation protocol used in the validation	

- ☒ (comprehensive list of documents attached clearly referenced)
- ☒ List of persons interviewed by DOE validation team during the validation process
- ☒ Any other documents. Please specify.
Please refer to list of documents attached
- ☒ Information on when and how the above validation report is made publicly available.
- ☐ Banking information on the payment of the non-reimbursable registration fee
- ☒ A statement signed by all project participants stipulating the modalities of communicating with the Executive Board and the secretariat in particular with regard to instructions regarding allocations of CERs at issuance

Executive Summary and Introduction, including

- **Description of the proposed CDM project activity**
- **Scope of validation process (include all documentation that has been reviewed and name persons that have been interviewed as part of the validation, as applicable)**
- **DOE Validation team (list of all persons involved in the validation, describing functions assumed in the validation)**

- **Description of the proposed CDM activity**

This report concerns the validation by SGS of the CDM Project for GHG Emission Reduction by Thermal Oxidation of HFC23 at the HCFC Plant of Gujarat Fluorochemicals Ltd (GFL). The project involves the installation of a thermal oxidation unit at the HCFC22 plant operated by GFL at their site in Gujarat, India. This unit will destroy the HFC23 which would otherwise be vented to atmosphere. The process will result in the emission of CO₂, both directly as a result of the destruction process and indirectly due to the use of LPG, steam, hydrated lime and caustic soda in the activity.

- **Scope of validation process**

The objectives of the validation exercise are to confirm that the project meets the necessary CDM criteria, that the project follows the approved methodology (AM0001) and that the proposals presented by GFL in the Project Design Document (PDD) will lead to a realistic determination of the quantities of HFC23 destroyed.

Scope of Assessment

The scope of the validation covers the baseline methodology involving a review of historical data (required to assess the performance of the plant before implementation of the project and hence to determine cut-off or limiting ratio of HFC23 generation: HCFC22 production and a study of the monitoring methodology and plan proposed for the assessment of the quantities of HFC23 destroyed and the emissions of greenhouse gases associated with the destruction process.

Documents reviewed

The following documents were reviewed as part of the scope of the activity:

- CDM Project Design Document "PROJECT FOR GHG EMISSION REDUCTION BY THERMAL OXIDATION OF HFC 23 at HCFC22 PLANT of GUJARAT FLUOROchemicals LIMITED (GFL)" prepared by Price Waterhouse Coopers, dated November 14 2003, updated 8 March 2004; final version June 23, 2004
- Host Country Letter of Approval dated 5 January 2004;
- Gujarat Pollution Control Board "No Objection Certificate" dated 15 March 2004; and
- CDM procedures under the Kyoto Protocol and Marrakech Accords..
- EIA Report reviewed during validation (not required under Indian legislation); confidential
- GFL Strategic Business Plan; confidential
- Ozone Depleting Substances (Regulation and Control) Rules, 2000, Government of India

Persons interviewed

The following persons were interviewed on issues relating to the project during the site visit:

Gujarat Fluorochemicals Ltd

- /1/ Deepak Asher – Vice President (Corporate Finance)
- /2/ V K Soni – Head of Projects
- /3/ Joseph Titus – Chief General Manager (Production)
- /4/ D K Sachdeva – Vice President (Operations)
- /5/ P S Parameswaran – General Manager (Technical)
- /6/ Rajeev Gupta – Dy General Manger (Projects)
- /7/ Akshay Parmar – General Manager (Engineering)

Ineos Fluor / ICI

- /8/ Pete Sleigh – European Regulatory Manager – Ineos Fluor UK
- /9/ Murali Duvvuri – Market Manager – ICI India

PriceWaterhouseCoopers

- /10/ Sameer K Singh

Other

- /11/ Lex de Jong, VROM, Govt of Netherlands (by telephone)

- **DOE Validation team**

The validation was undertaken by SGS using an Assessment Team of three people, two of whom visited the GFL site in India. John Miles acted as Team Leader and concentrated on the measurement and data processing issues, whilst Shivananda Shetty reviewed the local legislative and other requirements as well as studying the sampling and laboratory procedures. Marco van der Linden did not visit the GFL site but undertook the review of the project against CDM and other relevant criteria. The recommendation was then subjected to a Technical Review by Gareth Phillips.

Description of methodology for carrying out validation

- **Review of CDM-PDD and additional documentation attached to it**
- **Assessment against CDM requirements (e.g. by use of a validation protocol)**
- **Report of findings by the DOE, e.g. by use of type of findings (e.g. corrective action requests, clarifications or observations). Please explain the way findings are “labelled” during validation.**
- **Include statements or assessments in the section “Conclusions, final comments and validation opinion” below.**

The validation was performed as a desk review in combination with a site visit to the project site. The desk review consisted of an assessment of the PDD against the Approved Methodology, CDM requirements and other relevant criteria. The desk review was followed by a site visit to GFL Gujarat. Additional information was obtained from selected experts in the field.

A risk based audit approach has been adopted where issues of critical importance to the successful validation have been addressed in more detail.

- **Review of CDM-PDD and additional documentation attached to it**

The Project Design Document submitted by the Client was reviewed against the Approved Methodology, CDM requirements and other relevant criteria. Additional background documents related to the project design and baseline were also made available during the site visit to GFL. These documents were also reviewed.

- **Assessment against CDM requirements**

This approach has led to the development of a validation protocol (VP) which serves the following purposes:

- It organises, details and clarifies the requirements the project is expected to meet; and
- It ensures a transparent validation process where the validator will document how a particular requirement has been validated and the result of the validation.

The validation protocol was based on the template of the IETA / World Bank Validation and Verification Manual (<http://www.ieta.org/VVM/VVM.htm>).

- **Report of findings**

As an outcome of the validation process, Corrective Action Requests (CAR) are issued, where:

- i) mistakes have been made with a direct influence on project results;
- ii) determination protocol requirements have not been met; or
- iii) there is a risk that the project would not be accepted as a CDM project or that emission reductions will not be verified.

The term Clarification may be used where:

- iv) additional information is needed to fully clarify an issue.

Two different types of Corrective Action Requests are defined:

- **MAJOR CAR:** the validation is suspended or denied until the corrective action has been proven to be completed satisfactory;
- **MINOR CAR:** the validation is granted under the condition that the corrective action will be properly executed or implemented.

Corrective Action Requests (CARs) and Observations are raised in the draft validation protocol and the Project Developer is given the opportunity to “close” outstanding CARs and respond to Clarifications and Observations. Normally a time frame is attached to a CAR that reflects the importance of it. MAJOR CARs must be closed out before an unconditional validation report and

opinion can be issued. MINOR CARs may be closed out during the project implementation and verified during the first verification or surveillance visit. Observations must be addressed by the project developer prior to the first verification of avoided emissions. CARs may be closed out on the basis of documentation which will be verified during the first field visit.

Once a deadline for closure of a CAR has been passed, it may be escalated to a higher level – an outstanding MAJOR CAR may lead to suspension of the validation opinion; a MINOR CAR may be raised to a MAJOR CAR etc. and an Observation may be raised to a CAR. Outstanding Observations and CARs should be considered by the verification team at each subsequent verification assessment.

- **Close out and recommendation**

Following the closure of any outstanding issues and clarification, the Team Leader prepares a recommendation supported by the validation report, validation protocol and other documentation and objective evidence gathered by the team. The purpose of the report and protocol are to provide justification to support the Team Leader's recommendation.

- **Technical Review and Decision to issue an opinion**

The Final Report, Protocol, procedural documentation (such as contract review and skills justification) and other forms of objective evidence gathered during the engagement are sent to and Independent Decision Maker and Technical Reviewer who undertakes a review of the whole engagement and makes a decision whether or not to accept the Team Leader's recommendation.

- **Request for Review**

Three members of the CDM EB raised questions regarding the request for registration. These were considered by the Executive Board at its Seventeenth Meeting and the Executive Board agreed to carry out a review of the project activity based on the scope which included the following three issues:

- a) Clarification whether the host country approval dated 9th November, 2004 replaced the host country approval dated 5th January 2004
- b) Disclaimer contained in the validation opinion issued by SGS United Kingdom Ltd.'s
- c) The possible implications of HCFC 22 as a greenhouse gas

These were addressed in the following way:

- a) The Host Party issued a letter dated 30th December, 2004 confirming that the letter issued on 9th November 2004 replaced the earlier letter of approval dated 5th January 2004. The letter dated 9th November 2004 contained all the elements required of a Host Party letter.
- b) SGS United Kingdom Ltd. agreed to remove the disclaimer on the grounds that all of the data used to determine the baseline emissions had been verified during the site visit, with the exception of two numbers, that relate to the GWP of HFC 23 and its molecular weight, both of which are established by scientific study and widely reported.
- c) As regards the possible implications of HCFC22 being a greenhouse gas:
 - a. The PPs corrected the statement in the CDM-PDD which said that "HCFC22 is not a greenhouse gas" to "Though HCFC 22 is not listed in the Annex A of the Kyoto Protocol as a greenhouse gas, it is recognized as a greenhouse gas by the IPCC"
 - b. The PPs explained that the emissions of HCFC22 are not regulated by the Host Party. This fact was also verified by the validation team to be in line with the Montreal Protocol and the Ozone Depleting Substances (Regulation and Control) Rules, 2000,

Government of India.

- c. The PPs submitted the Strategic Business Plan (prepared prior to knowledge of CDM) upto 2010 which showed that the production of HCFC22 was planned to increase within the existing production capacity and in line with historical growth trends. This fact was also confirmed by the validation team.
- d. The PPs confirm that they would not claim any CERs relating to HCFC22 production in excess of its Strategic Business Plan, or actual HCFC22 production, whichever is lower
- e. The PPs submitted detailed information on Environment Impact of the proposed project activity

Explanation by the submitting designated operational entity of how it has taken due account of comments on validation requirements received, in accordance with the CDM modalities and procedures, from Parties, stakeholders and UNFCCC accredited non-governmental organizations;

- **Description of how and when the PDD was made publicly available**
- **Description of how comments were received and made publicly available**
- **Explanation of how due account has been taken of comments received**
- **Compilation of all comments received (Identify the submitter)**

- **Description of how and when the PDD was made publicly available**

SGS published the project documents on its website from 2003/12/01 until 2003/12/31 and public comments were invited through the UNFCCC website (refer to <http://cdm.unfccc.int/Validation/publicPDD/view.html?ProjectId=160945808326172952&OE=SGS-UKL>).

- **Description of how comments were received and made publicly available**

Comments could be submitted by email or fax. Comments are made publicly available through the SGS website

Compilation of all comments received

No comments were received within the 30 day stakeholder consultation period. One comment was received outside the 30 day period and therefore was not made publicly available. The nature of the comment did not require further action.

Comment number: n/a

Date received: n/a

Submitter: n/a

Comment: n/a

- **Explanation Of How Comments Have Been Taken Into Account**

Nothing to report.

Conclusions, final comments and validation opinion

- Provide conclusions on each requirement under paragraph 37 of the CDM modalities and procedures, describing how these requirements have been met. This shall include assessments and findings (e.g. corrective action requests, clarifications or observations) in relation to each requirement, including a confirmation that all issues raised have been addressed to the satisfaction of the DOE.
- Final comments and validation opinion

Project Design

The technology employed in the project is similar to that deployed by Ineos Fluor in their plant at Runcorn UK. This technology also forms the basis of the Approved Methodology (AM0001), which was developed for a similar project in Ulsan, Korea. The technology employed by Ineos Fluor is regarded as state of the art and its application in Runcorn is subject to stringent inspection for compliance with UK environmental regulations, and separately under the UK Emissions Trading Scheme. It is therefore reasonable to argue that the project employs current best practice and that, on the experience gained in Runcorn, the project will produce the maximum achievable reduction in GHG emissions. Both project duration and crediting time are clearly described in the PDD and are considered reasonable for this type of technology.

The project's contribution to sustainable development is achieved by increasing local incomes in the region of Gujarat in which the project will be based and by reducing vulnerability whilst increasing empowerment of vulnerable sections of society. In parallel GFL are committed to the improvement of natural resources in the vicinity of the project activity.

Baseline

The PDD describes the Baseline Methodology which is in conformance with the Approved Methodology (AM 0001). The key conclusions are summarised below, but further details are given in the Validation Protocol.

There are no regulations in India requiring the destruction of HFC23 and there is no market for the sale of HFC23 in India. It is therefore realistic to argue that the baseline condition should be that where all the HFC23 generated during HCFC22 production is being emitted to atmosphere.

It is accepted that the chloroform mass balance should be used to derive the cut-off ratio, which should be set at 2.9%, the lowest ratio achieved over the last three years. The revised PDD provides the justification for this approach.

It is not mandatory for GFL to install the thermal decomposition facility and, if installed, the facility will involve economic/financial commitment for GFL, consisting of capital and operating (recurring) costs without economic benefits besides the CERs. Therefore it is justified that the project activity itself is not a likely baseline scenario.

All the relevant GHG sources are identified, the system boundaries defined and the methods of determining the emissions are adequately described. Using these methods, estimates of emissions reductions using an illustrative production of HCFC22 of 10000MT per year show that emissions reductions due to the project will be in excess of 3 million tonnes CO₂e per year. The emission reductions will increase in future years in tandem with HCFC22 production in line with the company's business plan.

Monitoring Plan

It is considered that the Monitoring Plan will provide the necessary information to monitor and report emissions reductions due to the project. The protocol will allow the contribution from all sources to be identified in relation to the project operation and baseline factors.

The frequency of recording for parameters to be measured or calculated in connection with the project activity is specified in the PDD and ranges from monthly to yearly in line with the requirements of the Approved Methodology. Certain calibration checks and other tests on the installed instrumentation will be carried out more frequently than specified in the Approved Methodology, thereby providing improved accuracy and confidence in the measurements.

The responsibility within GFL for monitoring and reporting data relating to all project emissions and

emissions reductions is defined in the PDD. The measurement and monitoring procedures in use at the existing GFL plant in Gujarat will be extended to cover the new thermal oxidation unit and the QA manual will be modified accordingly.

Monitoring of Environmental Impacts is not explicit in the monitoring plan but is subject to environmental legislation and control by the Gujarat Pollution Control Board.

The PDD discusses the project's contribution to sustainable development. Whilst the monitoring plan does not currently identify indicators of sustainable development nor provide for monitoring of this contribution, GFL has agreed to identify such indicators and means of monitoring them, before the first verification. An Observation was raised to highlight this issue to the project developer and the verifier.

Calculation of GHG Emissions

The calculation methods applied to the determination of GHG emissions are explained in detail in the PDD. They follow the procedures laid down in the Approved Methodology.

All the relevant GHG sources are identified and accounted for in the PDD and the project boundaries are properly defined. The formulae and factors used for the determination of GHG emissions follow accepted practice and, where possible, conversion and other factors have been compared with those used in similar projects elsewhere. No significant differences were found.

The calculation of emissions reductions is based primarily on the measurement of HFC23 generated during the production of HCFC22. This measurement is based on the use of state of the art technology, ensuring that the measurement uncertainty will be as low as is practicably possible.

It was noted that although an allowance has been made for the emissions generated by the transport of sludge to landfill, no similar allowance had been made in the PDD or AM for the transport of LPG to the GFL site. GFL argued that these emissions could be ignored for the following reasons:

- The emissions will be smaller than those associated with sludge transport
- The emissions are only a very small part of the those generated by the project, which are in turn approx 0.04% of the reduction in emissions.
- The emissions will be outside the control of GFL and will be difficult to determine.

This argument is realistic and is therefore accepted.

Environmental Impacts

The PDD states that an Environmental Impact Assessment (EIA) for the proposed project has been undertaken. Existence of the report has been verified during the site visit and a copy made available for inspection subsequently.

Under Indian Law when a project is below Rs 50 crores and is not in the red category industry, approval of the EIA by the local pollution board is required. The project at GFL falls within this limit. A "Certificate of No Objection" by the Gujarat Pollution Control Board was provided dated 15 March 2004 (see Appendix C).

Comments by Local Stakeholders

GFL invited comments from local communities, shareholders, employees and the labour unions at meetings held at their Vadodara office on 13th June 2003. The responses from GFL are summarised below.

The local community representatives' concerns about the environmental impact of the project, GFL's role in conserving water resources and the employment potential of the project, were answered as follows:

- The project will conform to the best environmental performance standards. An EIA will also be conducted and a plan drawn up to ensure minimal adverse environmental impact.
- Most of the water required by the proposed system will be recycled, although about 15 m³/day will be drawn from external sources. GFL will continue their initiatives to improve water management in the region.
- Between 30 and 40 new jobs will be created by the Project and where possible these will be offered to local people.

The shareholders questions concerning the capital investment and its impact on GFL's bottom line;

potential alternative uses for HFC23; possible liabilities for past venting of HFC23; and the requirements for new employees or re-training existing employees were answered as follows:

- The project may have a positive impact on GFL's bottom line from the revenue received through the transfer of CERs.

Validation Opinion:

Validation Opinion Number 102154/01/05

SGS has performed a validation of the CDM project for GHG emissions reduction by thermal oxidation of HFC23 at Gujarat Fluorochemicals Limited, India. The Validation was performed on the basis of the UNFCCC criteria and host country criteria, as well as criteria given to provide for consistent project operations, monitoring and reporting.

The review of the project design documentation and the subsequent follow-up interviews have provided SGS with sufficient evidence to determine the fulfilment of the stated criteria. In our opinion, the project meets all relevant UNFCCC requirements for the CDM and all relevant host country criteria. The project will hence be recommended by SGS for registration with the UNFCCC.

SGS has received confirmation by the host Party that the project activity assists it in achieving sustainable development.

By destroying HFC23 (a potent greenhouse gas) generated during the production of HCFC22, the project results in reductions of greenhouse gas emissions that are real, measurable and give long-term benefits to the mitigation of climate change. An analysis of the investment and technological barriers demonstrates that the proposed project activity is not a likely baseline scenario. Emission reductions attributable to the project are hence additional to any that would occur in the absence of the project activity. Given that the project is implemented as designed, the project is likely to achieve the estimated amount of emission reductions.

Revised Validation Opinion Date 6th January 2005

The validation is based on the information made available to SGS and the engagement conditions detailed in the report. SGS has requested and obtained all the information needed, in its professional judgement, to reach the above opinion.

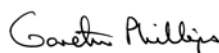
The DOE declares herewith that in undertaking the validation of this proposed CDM project activity it has no financial interest related to the proposed CDM project activity and that undertaking such a validation does not constitute a conflict of interest which is incompatible with the role of a DOE under the CDM.

By submitting this validation report, the DOE confirms that all validation requirements are met.

Name of authorized officer signing for the DOE

G. Phillips

Date and signature for the DOE



8th March 2005

Section below to be filled by UNFCCC secretariat

Date when the form is received at UNFCCC secretariat

Date at which the registration fee has been received		
Date at which registration shall be deemed final		
Date of request for review, if applicable		
Date and number of registration	Date	Number